

GPS and ZigBee Based Traffic Signal Preemption

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Abstract— It is very unsafe for Emergency Vehicle (EV) to wait near the traffic intersection for green signal. It needs to reach its destination as fast as possible. Even though emergency vehicles have higher priority on the road, but still at the traffic signal there is lag to give enough priority to pass through the signal. Sometimes they need to wait for the signal to favor them in a congested traffic or they may have to take part in dangerous movement which may result in collision. The better way to reduce the waiting time of such vehicles is by making the traffic signal green for providing easy pass through. So that the safety of other vehicles are also taken care. In this paper, whenever the emergency vehicle is approaching preemption messages are provided to the traffic signal unit. So these signals interrupt the normal operation of the traffic light and make it green for the emergency vehicle. It is a type of Vehicle-to-Infrastructure (V2I) communication. The proposed system uses GPS method to get the location data and employs ZigBee for communicating efficiently with the intersection module. There is no requirement of Line-of-Sight employed in infrared based preemption methods. Also eliminate the false trigger occurring in sound (siren) based method due to noise and fading. Using this approach, it is possible to serve multiple emergency vehicles reaching the traffic intersection according to their distance from the intersection and accurately taking the decision to turn the signal green for the particular vehicle. There is no involvement of driver of the EV in anyway, it is fully autonomous and it can be used as a part of ITS.

Keywords—ZigBee; V2I; ITS; GPS; OBU; XBEE; UART; GSM/GPRS; Preemption

I. INTRODUCTION

Due to urbanization and increased in use of private vehicles roads are getting congested. It is very common in urban area that vehicles are waiting for their turn to pass through the signal intersection. There are many methods to control the traffic light. Sometimes it is based on the time of the day and it may be adaptive control which may depend upon the vehicle density. Anyway both will help in reducing the wait time of vehicles depending upon their respective implementation techniques. Most of the traffic light control unit not considering some uncertain condition like some accidents happened, or some emergency vehicle need to pass through the signal.

Even though the occurrence of such an emergency is less but there is a need to consider such a type of situation which will add intelligence to the current flow of operation of the signal. Then there is a question regarding management of these kinds of situations. One way is to control the light by detecting this kind of situation. That is preemption of traffic signal. There are many methods for detection of emergency situation such as ambulance or fire engine needs to pass through the signal. These methods main purpose is safely allowing the path for EV and reduce the time delay occurring at the junction. This will save lives and other damages caused on property. And also since vehicles need not wait for much time so reduction in emission. So the main objective of all these methods is same but their implementation or technologies they are using have its own limitation.

In optical based emergency vehicle detection system the main requirement is line of sight between the detectors which may not possible at all time. Some objects will come in between or it may be due to bad weather conditions. In Acoustic based preemption system emergency vehicle's siren is used for vehicle approach. In this scheme falls triggering may occur due to acoustic reflection due to building or some other large container vehicles.



Fig. 1. Emergency vehicle preempting traffic light

The data from the EV is processed by the intersection unit and make a decision upon that. There is no involvement of driver for preemption. The hardware at OBU is implemented by using Arduino, GPS module and XBEE module. The intersection unit hardware is implemented using Arduino and XBEE module.

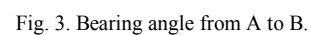
The proposed system is going to provide a simple and optimum solution to the main problem associated with the different pre-emption systems. Mainly in GPS based system which employs GSM/GPRS communication for transmitting the location details of the emergency vehicles may get delayed as mention in[1].This may be due to some connectivity problem associated with mobile communication network. In this system there are mainly the following problems are considered and resolved in system implementation:

- a. Delivering Vehicle information to the intersection unit without time delay compared to the GSM/GPRS methods. Vehicle information is transmitted using ZigBee transmission.
- b. Preemption point criteria – Preemption point is the distance between traffic intersection unit and the EV prompting the preemption request. This distance can be varied according to the traffic conditions at the particular intersection. In a heavy traffic junction this distance can be reduced compared to low density traffic intersection.
- c. The wait time of other vehicles can be decreased by considering the above criteria step b.
- d. It will also serve multiple vehicles according to their distance from the intersection. So the emergency vehicles near to the intersection will be served first.
- e. It will also update the other vehicles update and preemption request in a buffer and will serve those vehicle which is nearby as mentioned in step d.
- f. During this updates it will check the preempted vehicles updated position information and it will check that particular vehicle is crossed the junction. This will further reduce the wait time of other vehicle during the preemption time period.

The system depending upon the global positioning system GPS [9] situated in the vehicle. For GPS data can be from a dedicated GPS module or from the mobile phone. The reliable communication is provided by ZigBee radio communication technologies. Mainly ZigBee transmitter located at the vehicle and the receiver module at intersection provide the communication link.

Before preemption request the unit is in normal mode of operation, it will receive the message from EV and once preemption is required it will move into the preemption phase for that particular vehicle. In this phase it will check the particular vehicle is crossed the junction and timeout which ever happens first the unit phase return to normal mode of operation. Fig 2 depicts these transitions.

It will tell the direction the vehicle need to move to reach a point. Measured in angle clockwise from the North direction. Fig. 3 shows the direction of point B from point A. This angle is used for finding the road from which the EV is approaching the junction.



III. HARDWARE IMPLEMENTATION OF THE SYSTEM

A traffic control preemption system uses data received from a Global Positioning System (GPS) to determine whether a vehicle requesting a preemption is within an allowed path of that particular intersection. The vehicle unit start capturing these data when the driver require/trigger preemption. These data are transmitted to the intersection module. Further processing is done based on these data and decision to preempt the signal is made. This will inform the main traffic light control to interrupt its operation for EV to pass through. After the vehicle clear the junction normal operation of the traffic light is maintained.

A. On-Board-Transmitting module

The vehicular unit continuously receive the GPS information from the GPS module connected to it and broadcast its location information using Xbee module. These GPS and Xbee modules are connected to a micro controller for processing the data coming from GPS and for sending. They are connected through the serial port of the controller. Arduino microcontroller is used.

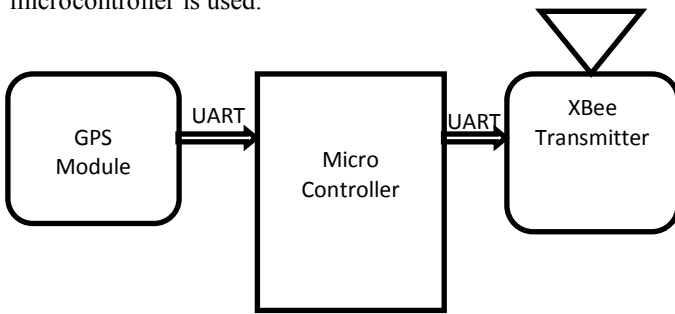


Fig. 4. Block diagram of Emergency Vehicle on board unit

B. Message format

There are two types of message format involved in this system implementation; *NMEA Sentence*- these are the GPS string obtained from the GPS module mentioned in [5]. *Preemption Message*- this is the message passing between transmitter at the on-board unit and traffic intersection receiver module. It consists of particular EV ID, its current latitude, longitude, speed and heading direction obtained from GPS.

C. Traffic Intersection module

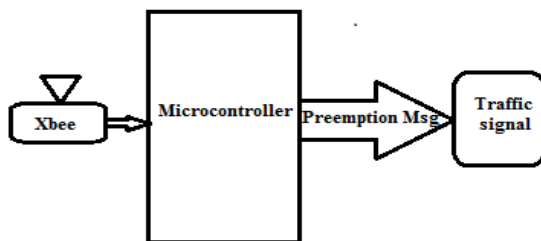


Fig. 5. Block diagram of Traffic Intersection module

Every traffic intersection consists of Traffic Intersection Unit. This unit is used for taking decision to change the normal operation of the traffic signal. It consists of XBee receiver which will receive the location related information of EV in its vicinity. It is connected to Arduino microcontroller which will process these information. XBee is connected to the controller through UART serial interface. If the processed data require preemption of the traffic light it will give indication for preemption.

IV. ALGORITHM AND MAIN FLOW CHART AT THE TRAFFIC INTERSECTION.

1. Receive the EV Preemption Message.
2. Calculate the distance between EV and Traffic intersection.
3. If the distance is less than Preemption point and currently Traffic signal is in Normal mode of operation Preempt the signal. If the vehicle is currently in preempting state for some other early arrived emergency vehicles near the intersection, store it in a waiting list.
4. If the distance calculated in step 3 is greater than preemption point and if it is less than store distance, store these vehicle information also in a list.
5. If the distance calculated in step 3 is more than store distance discard those messages because vehicle is far from the junction.
6. In step 3 if Preemption is initiated, find the direction of from which vehicle is approaching the junction. And TURN ON the GREEN signal for this direction for a fixed time duration and follow step 1 and step2.
7. Check the vehicle cross during this reception by receiving the vehicle's update during this reception.
8. If vehicle is crossed the junction preemption for that vehicle is completed and resume the normal operation of the traffic light. Otherwise preempt the signal till timeout.
9. Once the preemption is completed for one vehicle find the next nearby vehicle which requires the preemption from the stored waiting list and follow step 3.
10. If all the stored vehicles are fully processed from the list and repeat from step 1.

The flow diagram of the intersection unit (Preemption-unit) is shown in Fig 6. It is in a continuous loop which checks for any Preemption-message from EV. Once it receives the Preemption Message it will start processing and take decision to preempt the signal or not. Once it is in preempting state it will check for that vehicle crossed the junction or not depending upon the further message received from that vehicle or it will wait till timeout occurs. Normally timeout will be given according to the Preemption-point. It can vary from 10-150 sec, also depends upon the road topology and traffic density.

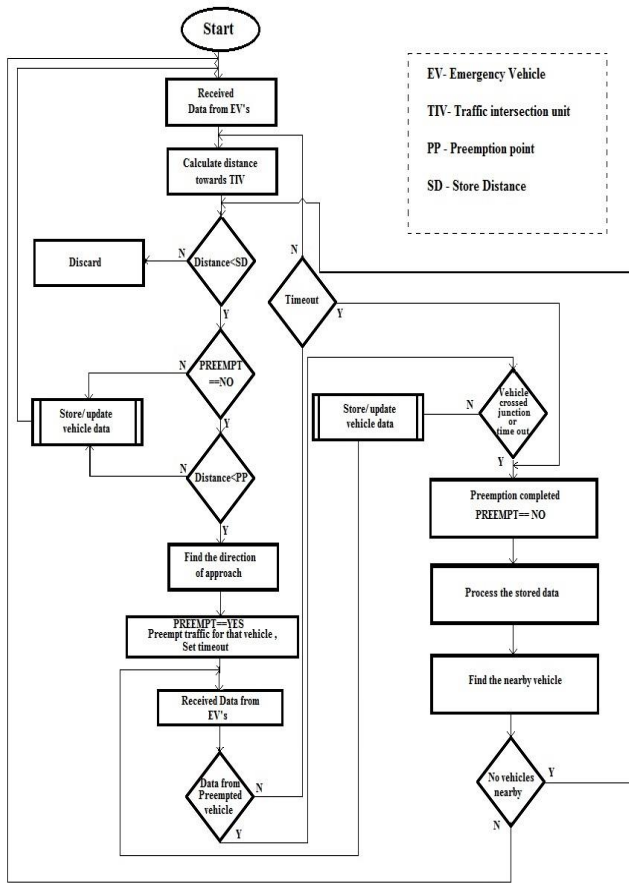


Fig. 6. Main flow diagram at intersection unit

V. RESULT

The hardware setup for On-Board-Unit which is located at each EV for sending the GPS update value of that vehicle is shown in Fig 7.

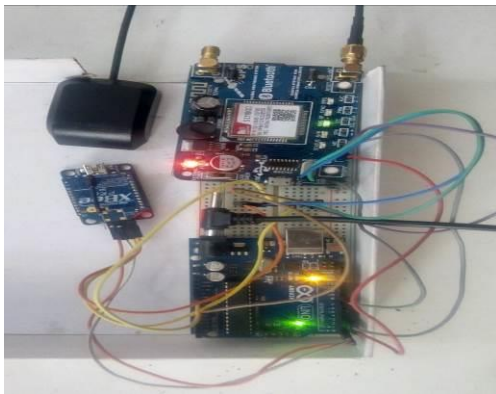


Fig. 7. Emergency vehicle on board unit setup

The message output from the OBU is a Preemption Message which consists of vehicle ID and the GPS values read from the GPS module. Once GPS acquire the position information it will start sending if preemption is needed.

The message delivery count depends on the ZigBee transmit and receive rate mentioned [4]. The message transmission count also depends on the number of EV count transmitting at the same time. So the Preemption Message frequency is fixed by considering the EV count and the ZigBee reception rate at the Intersection unit.

The module that is doing the main part of this system is Traffic-Intersection unit. Fig.8 shows the Intersection unit which receives the Preemption Messages from different vehicles and decides whether to preempt the traffic signal or not.

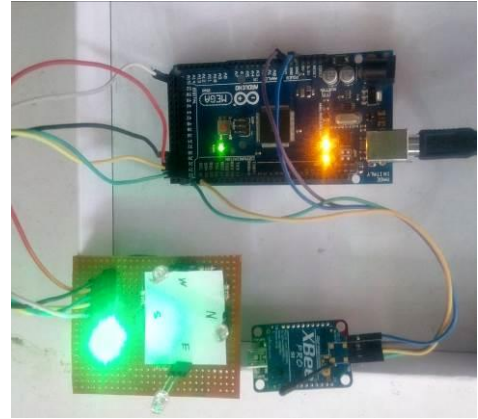


Fig. 8. Traffic intersection module setup

This unit will decide the preempting direction and TURN ON the green light as shown in Fig 8. from which the vehicle is approaching the intersection. Also the Intersection unit's Logging information shows about which all processing are going inside are shown in Fig 9. The logs are taken from serial monitor from Arduino IDE since Arduino controller is used at Intersection unit.

Fig. 9. Logging information from traffic preemption module

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COM3 (Arduino/Genuino Mega or Mega 2560)

Compare direction=1
Exec time for check direction=862
PREEMPTING THE TRAFFIC ----W----- PREEMPTING DIRECTION
PREEMPTING VEH----- PREEMPTING VEHICLE ID
1234-----
Inside store2_prem_vehi_info:search no=0
W
Course=14.64Execution time preme inside fun=1055
Execution time preempt=1100
Executiontime=1461
5
5
1302
30000
Switch2-preempt
inside recv_msg
length=34
1234,12.917750,77.681600,5,185.235
First item in string: 1234
Second item in string: 12.917750
Third item in string: 77.681600
Fourth item in string: 5
Fifth item in string: 185.23
12.917751
77.681594
5.000000
185.229995
1234
Distance Towards traffic light=0.009815-----DISTANCE TOWARDS JUNCTION
Stage c- Preempt-Preemption Vehicle=YES
Inside Check veh cross
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The delay associated with processing and taking the decision for EV is given in bellow table. In this paper mapping of the road points is done on the possible approach for better prediction accuracy. The main cause of delay is in comparing the road points with current vehicle location .This comparison is for double checking the direction of vehicle about the approach towards the particular intersection. This delay is depends upon the road topology and can be minimized till up to best case as mentioned in the Fig. 10.

Number of mapped road points in each of 4 direction	Worst Case Execution Time (m.sec)		
	Finding Direction	Finding Vehicle is crossed or not	Execution time for Preemption (which includes time for finding direction)
10	540	647	776
12	604	712	841
15	702	809	893
17	766	874	958
20	862	968	1055
Best Case Execution Time (m.sec)			
	165	27	357

Fig. 10. Latencies associated with preemption module.

VI. RESULT COMPARISON AND CONCLUSION

The main design goal of this paper is to reduce the time delay and process the EV as fast as possible without any miss prediction of direction. Since ZigBee handles a reliable message delivery in its range of operation, the delay caused by GSM method by [1] is avoided. The time taken to preempt the signal in worst case is 1sec. The false decision problem in

acoustic preemption method on EV is eliminated using this method. Extra detectors and sensors circuit used in other methods to find the vehicle position or to detecting the vehicle entering the preemption point is also avoided here.This implementation is allowing to vary Preemption point without any extra circuitry and sensors. It is possible to customize and maintain the system easily as less components are involved in the system.

References

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